

Fermat's little theorem

In number theory, Fermat's little theorem states that if p is a prime number, then for any integer a , the number $a^p - a$ is an integer multiple of p . In the notation of modular arithmetic, this is expressed as

$$a^p \equiv a \pmod{p}$$

For example, if $a = 2$ and $p = 7$, then $2^7 = 128$, and $128 - 2 = 126 = 7 * 18$ is an integer multiple of 7.

If a is not divisible by p , that is, if a is co-prime to p , then Fermat's little theorem is equivalent to the statement that $a^{(p-1)} - 1$ is an integer multiple of p , or in symbols.

$$a^{p-1} \equiv 1 \pmod{p}$$

For example, if $a = 2$ and $p = 7$, then $2^6 = 64$, and $64 - 1 = 63 = 7 * 9$ is a multiple of 7.

Proof.

Start by listing the first $p-1$ positive multiples of a :

$$a, 2a, 3a, \dots (p-1)a$$

Suppose that ra and sa are the same modulo p , then we have $r \equiv s \pmod{p}$, so the $p-1$ multiples of a above are distinct and nonzero; that is, they must be congruent to $1, 2, 3, \dots, p-1$ in some order.

Multiply all these congruences together and we find

$$a (2a) (3a) \dots ((p-1)a) \equiv 1.2.3.....(p-1) \pmod{p}$$

which is, $a^{(p-1)(p-1)!} \equiv (p-1)! \pmod{p}$. Divide both side by $(p-1)!$ to complete the proof.

Corollary.

Let p be a prime and a any integer, then $a^p \equiv a \pmod{p}$.

Proof.

The result is trivial (both sides are zero) if p divides a . If p does not divide a , then we need only multiply the congruence in Fermat's Little Theorem by a to complete the proof.

Modular Multiplicative Inverse

Problem

Given value of A and M , find the value of X such that $AX \equiv 1 \pmod{M}$.

For example, if $A = 2$ and $M = 3$, then $X = 2$, since $2 \times 2 = 4 \equiv 1 \pmod{3}$.

We can rewrite the above equation to this:

$$\begin{aligned} AX &\equiv 1 \pmod{M} \\ X &\equiv 1/A \pmod{M} \\ X &\equiv A^{-1} \pmod{M} \end{aligned}$$

Hence, the value X is known as Modular Multiplicative Inverse of A with respect to M .

How to Find Modular Inverse?

First we have to determine whether Modular Inverse even exists for given A and M before we jump to finding the solution. Modular Inverse doesn't exist for every pair of given value.

Existence of Modular Inverse

Modular Inverse of A with respect to M , that is, $X = A^{-1} \pmod{M}$ exists, if and only if A and M are co-prime.

Why is that?

$$\begin{aligned} AX &\equiv 1 \pmod{M} \\ AX - 1 &\equiv 0 \pmod{M} \end{aligned}$$

Therefore, M divides $AX - 1$. Since M divides $AX - 1$, then a divisor of M will also divide $AX - 1$. Now suppose, A and M are not co-prime.

Let D be a number greater than 1 which divides both A and M . So, D will divide $AX - 1$. Since D already divides A , D must divide 1.

But this is not possible. Therefore, the equation is unsolvable when A and M are not co-prime.

From here on, we will assume that A and M are co-prime unless state otherwise

Using Fermat's Little Theorem

It stated that, if A and M are co-prime and M is a prime, then, $A^{(M-1)} \equiv 1 \pmod{M}$. We can use this equation to find the modular inverse.

$$\begin{aligned} A^{(M-1)} &\equiv 1 \pmod{M} \text{ (Divide both side by } A) \\ A^{(M-2)} &\equiv 1/A \pmod{M} \\ A^{(M-2)} &\equiv A^{-1} \pmod{M} \end{aligned}$$

Therefore, when M is prime, we can find modular inverse by calculating the value of $A^{(M-2)}$. How do we calculate this? Using Modular Exponentiation.

This is the easiest method, but it doesn't work for non-prime M .

Why Do We Need Modular Inverse?

We need Modular Inverse to handle division during Modular Arithmetic. Suppose we are trying to find the value of the following equations:

$4/2 \% 3$ - This is simple. We just simplify the equation and apply normal modular operation. That is, it becomes $4/2 \% 3 = 2 \% 3 = 2$.

Then what happens when we try to do same with $12/9 \% 5$? First we simply, $12/9 \% 5 = 4/3 \% 5$. Now we are facing an irreducible fraction. Should we simply perform the modular operation with numerator and denominator?

That doesn't help since both of them are smaller than 5.

This is where Modular Inverse comes to the rescue. Let us solve the equation $X \equiv 3^{-1} \pmod{5}$. How do we find the value of X ? We will see that on the later part of the post. For now, just assume that we know the value of X .

Now, we can rewrite the above equation in the following manner:

$$\begin{aligned} 12/9 \% 5 \\ 4/3 \% 5 \\ (4 \times 3^{-1}) \% 5 \\ ((4 \% 5) \times (3^{-1} \% 5)) \% 5 \\ \therefore 4X \% 5 \end{aligned}$$

So, now we can easily find the value of $A/B \% M$ by simply calculating the value of $(A \times B^{-1}) \% M$.